AI Image Generation using Diffusion Model

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An Interim Report submitted in fulfilment of the requirements for the degree of

Bachelor of Engineering in Electronic and Information Engineering

Department of Electrical and Electronic Engineering
Imperial College London

2024

Abstract

Denoising Diffusion Probabilistic Models (DDPMs) have rapidly become one of the leading families of generative models, offering a stability of training that Generative Adversarial Networks (GANs) have historically struggled to achieve. Within a DDPM the diffusion process is fixed and the only learned component is the network that predicts the noise added at each step, so the choice of that network governs the quality of the images produced. This project asks whether combining residual learning with the UNet backbone conventionally used for that purpose yields a measurable improvement.

A residual UNet (ResUNet) has been implemented in PyTorch, in which every encoder and decoder stage is constructed from residual blocks rather than plain convolutional stacks, while skip connections carry encoder features across to the corresponding decoder stage. It is trained directly on the standard DDPM objective: a timestep is sampled uniformly, the noised image is formed in closed form from the cumulative noise schedule, and the network is optimised to predict the Gaussian noise that was added.

The network is assessed against two baselines under matched conditions. The first removes the identity shortcut from each block and is otherwise identical, so that the two networks have exactly the same parameter count and the comparison isolates residual learning as the single variable. The second is the denoising architecture of the original DDPM, with four resolution stages and self-attention, reduced in width so that its parameter count agrees with the proposed network to within a few per cent. All three are trained on CIFAR-10 at 32×32 and on CelebA at 64×64, sharing the diffusion schedule, the optimiser, the number of epochs and the random seed, and sampling from the same initial noise.

Quality is judged by the training objective, by a structural similarity diagnostic that measures how completely a noised test image is recovered, and by inspection of samples drawn from the full reverse process. The distribution-level metrics identified in the project specification, namely the Fréchet Inception Distance, the Inception Score and the Negative Log-Likelihood, together with the comparison against Variational Autoencoder and GAN baselines, remain to be carried out.

This report establishes the theoretical background for that programme. It reviews the literature on diffusion probabilistic models and the stochastic differential equation formulation of the diffusion process, examines Variational Autoencoders and GANs as competing generative approaches, and sets out the residual learning principles on which the proposed denoising network is founded.

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1 Project Specification

1.1 Introduction

This research project is centered on examining the capabilities of Denoising Diffusion Probabilistic Model (DDPM) in the field of image generation. The focus of this study is the generation of natural images, evaluated on two datasets of differing resolution. The first is CIFAR-10, the standard benchmark against which diffusion models are reported, comprising 32×32 RGB images drawn from ten object classes; retaining it allows the results obtained here to be placed alongside the published literature. The second is CelebA, a large-scale collection of aligned face photographs, from which a centre crop is taken and resized to 64×64. The higher resolution is included because architectural differences between denoising networks are difficult to observe at 32×32, where the spatial extent is exhausted after only two downsampling stages; at 64×64 a third stage becomes available and the contribution of network depth and of the skip connections is visible both in the metrics and in the generated samples. So that the two settings remain comparable in training cost, a 50,000-image subset of CelebA is used, matching the size of the CIFAR-10 training split. The project aims to not only develop and train a DDPM model but also to enrich the research with comprehensive analytical work, simulations, and the development of dedicated software tools. This approach intends to merge theoretical insights with practical applications, contributing significantly to the understanding and advancement of generative models in digital image processing.

In this research project, I will showcase the generative capabilities of diffusion models by comparing them to established competitors in the field of image generation: Generative Adversarial Network (GANs) and Variational Auto Encoder (VAEs). This study aims to provide a thorough comparative analysis, highlighting the strengths and limitations of each model type. Additionally, the project will explore an alternative approach by combining residual learning with the UNet backbone that is conventionally adopted for the denoising network, giving a residual UNet (ResUNet) in which each encoder and decoder stage is constructed from residual blocks rather than plain convolutional stacks. This investigation into different architectural frameworks is expected to yield insightful results regarding efficiency, effectiveness, and the quality of image generation, thus contributing to the broader discourse on machine learning architectures in generative applications.

1.1.1 Software

This project involves the implementation of a Denoising Diffusion Probabilistic Model (DDPM) with an emphasis on the denoising process, for which a residual UNet (ResUNet) serves as the noise-prediction network. Training follows the standard DDPM objective: a timestep is drawn uniformly for each example in a batch, the corresponding noisy image is formed in closed form from the cumulative noise schedule, and the network is optimised to predict the Gaussian noise that was added. The quantity minimised is therefore a noise-prediction error, not a reconstruction error. A key component of the framework is the data handling routine, which caches CIFAR-10 and CelebA as tensors at the required resolution so that an identical training loop applies to either dataset without modification. Dedicated classes are provided for training, sampling and evaluation, and the diffusion process is implemented separately from the denoising network so that the architecture under test can be exchanged without altering the generative model itself. This separation is what makes the controlled comparison of Section 1.1.2 possible.

1.1.2 Simulation

The experimental programme holds the diffusion framework fixed and varies only the denoising network, so that any difference in performance can be attributed to the architecture rather than to the generative model. Three networks are trained on each of the two datasets, giving six runs in total.

- **ResUNet** is the proposed network, in which every encoder and decoder stage is built from residual blocks.
- **Plain UNet** is identical in every respect except that the identity shortcut is removed from each block. A shortcut adds no parameters, so the two networks have exactly the same parameter count and computational cost; this comparison therefore isolates residual learning as the single variable.
- **DDPM UNet** is the denoising architecture of Ho et al., with four resolution stages, self-attention at 16×16 and a skip connection taken from every residual block. Its base channel width is reduced so that its parameter count falls within a few per cent of the proposed network. Without that adjustment the published configuration would be an order of magnitude larger, and any difference in quality would reflect capacity rather than design.

All six runs share the number of diffusion steps, the noise schedule, the training objective, the optimiser and its learning rate, the number of epochs, the data augmentation and the random seed; sampling for every model begins from the same initial noise. Evaluation is carried out on three levels. The restoration diagnostics are computed over a thousand held-out test images at six separate noise levels; the distribution-level metrics require a far larger sample and are computed from two thousand images drawn through the complete reverse process for each model; and the memorisation check compares every generated sample against the entire training set. Analysing the models under these matched conditions is what allows patterns and variances in the diffusion process to be attributed to the architecture, and provides the basis on which the proposed network is judged.

1.1.3 Analytical Work

Throughout the project, a detailed analysis will be performed to document the functionality and improvements made at each stage. This documentation will emphasize the advancements achieved during the development process. By maintaining a rigorous and structured analysis, the project will capture incremental enhancements, troubleshooting steps, and key decisions that significantly impact model performance. This approach ensures that all aspects of the system—from the initial design to the final implementation—are well-documented, providing a clear record of the developmental journey. The documentation will not only serve as a valuable resource for assessing the project's success but also as a blueprint for future projects aiming to build upon this work. Highlighting these advancements will underscore the technical progress and innovative solutions developed during the project, showcasing the evolution of the model from concept to execution.

No single number captures the performance of a generative model, and the measurements reported here fall into three groups that answer different questions. The first group asks how faithfully a corrupted image is restored, and comprises the Structural Similarity Index Measure (SSIM) and the Peak Signal-to-Noise Ratio (PSNR). The second asks how closely the distribution of generated images resembles the distribution of real ones, and comprises the Fréchet Inception Distance (FID) and the Inception Score (IS). The third is not a quality measure at all

but a validity check: a nearest-neighbour comparison against the training set, which establishes that the model is generating rather than reproducing.

- **Structural Similarity Index Measure** compares two images in terms of luminance, contrast and structural information. It requires a matched reference image, and is therefore applied to the reconstruction of held-out test images rather than to unconditional samples: a test image is corrupted to a chosen timestep and recovered in a single step, and the recovered image is compared with the original. A higher value indicates a more faithful restoration.

- **Peak Signal-to-Noise Ratio** is computed on the same pairs and reports the mean squared error on a logarithmic scale, in decibels. It is reported alongside SSIM because the two quantify different things: SSIM responds to structural agreement, PSNR to pixel-wise error. Where the two disagree, the disagreement is itself informative, since it indicates that one network is more accurate in absolute terms while the other preserves structure better.

- **Fréchet Inception Distance** quantifies the distance between the feature vectors of real images and generated images calculated by a pre-trained Inception network. Unlike the two measures above it needs no reference image, so it applies to samples drawn from noise and is therefore the primary measure of generative quality in this work. A lower FID indicates a closer match between the generated and real distributions.

- **Inception Score** evaluates the clarity and diversity of generated images from the same samples. A higher IS suggests that the model produces distinct and confidently classified images, although the measure is known to reward confident classification even when the images are not realistic, and it is therefore read alongside FID rather than on its own.

- **Nearest-neighbour distance** locates, for every generated sample, the closest image in the training set under a pixel-space L_2 metric. A distance on its own cannot be interpreted, so the same measurement is taken for held-out real images that the model never saw; if the generated samples are no closer to the training set than genuine unseen images are, there is no evidence that the model is reproducing its training data.

Negative Log-Likelihood was considered and deliberately excluded. Obtaining it for a diffusion model requires evaluating the full variational bound rather than the simplified training objective, and the bound is dominated by the reverse-process variances. The implementation used here follows the original formulation in fixing those variances to the posterior values rather than learning them, so the resulting likelihood would report a property of that fixed choice more than a property of the denoising network under test. Reporting it would add a number without adding evidence about the question this project asks.

Taken together the three groups support different claims. The restoration measures compare the networks as denoisers, which is what they are trained to be; the distribution measures compare them as generators, which is what they are ultimately for; and the memorisation check establishes that the second claim is meaningful. Reporting all three guards against the failure mode of selecting whichever single metric happens to favour the proposed architecture.

2 Background

2.1 Review of Literature

Diffusion probabilistic models have emerged as a powerful class of generative models that have significantly advanced the field of image generation. These models operate by gradually transforming a distribution of random noise into a distribution of structured images over many iterative steps, effectively reversing the diffusion process. This methodology allows for the generation of highly detailed and coherent images across a range of domains and applications.

One of the most notable achievements of diffusion probabilistic models is their ability to produce images that rival the quality of those generated by other leading techniques, such as Generative Adversarial Networks (GANs). Studies have shown that diffusion models can achieve superior fidelity and diversity in generated images, often demonstrating fewer artifacts than those commonly seen in GAN-produced images.

In terms of specific accomplishments, diffusion models have excelled in tasks such as high-resolution image synthesis, text-to-image generation, and image-to-image translation. For instance, they have been successfully applied to create photorealistic images of landscapes and portraits from textual descriptions, showcasing their versatility and robustness. Additionally, these models have proven capable of handling complex conditioning information, making them particularly useful for tasks requiring detailed customization of the generated content.

The theoretical underpinnings of diffusion models also contribute to their appeal. Unlike GANs, which can be challenging to train due to issues like mode collapse, diffusion models offer a more stable training process. This stability stems from their foundation in well-understood stochastic differential equations, providing a clear pathway for systematic improvements and adaptations.

Overall, diffusion probabilistic models represent a significant step forward in the field of image generation, combining high-quality output with robust and stable training methodologies. Their continuing development promises further enhancements in generating detailed, diverse, and customizable visual content, making them a central focus of current research in artificial intelligence and computer vision.

The ResNet architecture, short for Residual Network, is a groundbreaking innovation in the field of deep learning that was introduced by Kaiming He et al., titled "Deep Residual Learning for Image Recognition," [1]. This architecture has since been widely adopted for various applications across computer vision tasks due to its powerful and efficient design.

ResNet's primary innovation lies in its use of residual blocks (Figure 2.1). These blocks allow the network to skip one or more layers through the use of shortcut connections, or skip connections, that perform identity mapping. By adding the output from an earlier layer to a later

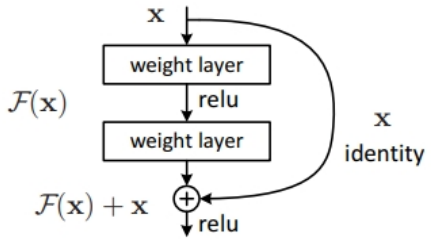


Figure 2.1: Residual Learning: a building block [1]

layer, these connections help to mitigate the vanishing gradient problem that plagues deep neural networks. As a result, ResNet can effectively train much deeper networks than was previously possible.

The standard implementations of ResNet include variants with different depths, typically ranging from 18 layers to 152 layers. ResNet-50, ResNet-101, and ResNet-152 are among the most popular configurations, with the number indicating the count of layers. These models have shown remarkable success in benchmark tasks such as image classification, detection, and segmentation on challenging datasets like ImageNet and COCO.

In terms of performance, ResNet models have dramatically reduced error rates in classification tasks compared to their predecessors. For instance, ResNet-50 and ResNet-101 have become go-to models for many researchers due to their balance of depth and computational efficiency, providing excellent accuracy without the exponential increase in computational cost usually associated with deeper models.

Moreover, the adaptability of ResNet is one of its standout features. It has been successfully applied in settings beyond image recognition, including audio signal processing and reinforcement learning. The architecture's ability to be modified and extended makes it a versatile tool for a wide range of applications in machine learning.

Overall, the ResNet architecture has proven to be a significant advancement in neural network design, enabling deeper and more powerful networks that can handle the increasing complexity of tasks across various domains. Its ongoing influence in research and application underscores its critical role in the evolution of deep learning technologies.

Training diffusion probabilistic models presents a unique set of challenges that are both computational and theoretical in nature. These models operate by learning to reverse a diffusion process, gradually converting random noise into a structured data distribution over many iterative steps. While the results can be impressive, achieving them is far from straightforward.

One of the most significant hurdles in training diffusion models is their high computational cost. Each training iteration involves simulating the reverse diffusion process across multiple timesteps, which can be computationally intensive especially as the number of timesteps increases. This makes training such models resource-heavy, requiring powerful hardware and often resulting in long training times. Researchers have explored methods such as subsampling timesteps during training to reduce computational overhead. This approach is discussed in "Improved Denoising Diffusion Probabilistic Models" by Alex Nichol and Prafulla Dhariwal [2], where they introduce techniques to selectively skip timesteps without compromising the model's performance.

The performance of diffusion models heavily relies on the precise tuning of hyperparameters, including the learning rate, the number of diffusion steps, and the noise schedule. Determining the optimal settings for these parameters can be a trial-and-error process that requires extensive experimentation and computational resources.

Although diffusion models are generally more stable during training compared to other generative models like GANs, they are not completely free from stability issues. Problems can arise from numerical instabilities during the simulation of the reverse diffusion process, particularly when dealing with very low probability events in high-dimensional spaces.

Diffusion models are often applied to high-dimensional data such as images, which can complicate the training process. The curse of dimensionality can exacerbate the challenges in modeling the data distribution accurately, requiring more sophisticated techniques and deeper networks. Techniques such as principal component analysis (PCA) and autoencoders have been used to reduce the dimensionality of data before training diffusion models. This can simplify the learning process and improve the manageability of high-dimensional data.

Measuring the performance of diffusion models can also be challenging. Traditional metrics

like log-likelihood are not always informative for image quality in visual tasks. Alternatives like Inception Score or Fréchet Inception Distance are commonly used, but they have their limitations and may not fully capture the model's capabilities in generating diverse and realistic images.

A practical challenge in training diffusion models for image generation is balancing the level of detail in the generated images with the diversity of the output. Overtraining on specific features can lead to loss of diversity (mode collapse), while too much randomness can reduce the realism of the generated images.

Despite these challenges, diffusion models are continuously being refined and improved, making them increasingly viable for a range of applications. Advances in computational resources, algorithmic optimizations, and better understanding of the underlying processes are helping to mitigate these difficulties, broadening the practical use of diffusion models in the field of generative modeling.

Exploring Stochastic Differential Equations (SDEs) as a modeling approach for the diffusion process has become a significant area of research in the development of diffusion probabilistic models. SDEs are used to provide a mathematical framework for modeling the continuous dynamics of the random variables involved in the diffusion process. This approach has several advantages, particularly in terms of flexibility, precision, and the ability to model complex stochastic behaviors.

Stochastic Differential Equations are used to model systems that exhibit random behavior and are influenced by continuous stochastic noise. In the context of diffusion models, SDEs describe how an image (or another data type) gradually transitions from a meaningful configuration to a state of pure noise—and vice versa—over continuous time. This is fundamentally different from the discrete step-wise approach seen in earlier diffusion models, where the process was modeled as a sequence of discrete steps.

In diffusion probabilistic models, the forward process is modeled as adding noise over time until the data becomes indistinguishable from Gaussian noise. The reverse process, which is of primary interest, involves learning the reverse of this noise-adding process. By employing SDEs, researchers can model this reverse process more naturally and fluidly as it provides a continuous-time framework, which can capture the nuances of the noise reduction at any point in time.

SDEs allow the diffusion process to be defined in continuous time, giving a finer control over the dynamics of the process. This can lead to more precise and controlled generation of the data, as the model can potentially make adjustments at any moment in the continuous timeline. The use of SDEs enables the modeling of more complex dynamics that cannot be captured by simple discrete transitions. This is particularly useful in applications where the data exhibits intricate stochastic behaviors. It is also possible to implement more efficient sampling algorithms that leverage the continuous nature of the process. For instance, adaptive step-size solvers can be used to speed up the reverse diffusion process without loss of quality.

The seminal work by Yang Song et al. in their paper "Score-Based Generative Modeling through Stochastic Differential Equations" published at ICLR 2021 [3], is a notable contribution to this area, and the correspondence between the forward and reverse-time processes that it formalises is illustrated in Figure 2.2. They propose a framework that uses score-based methods and SDEs to generate samples. By defining the generative model as a solution to an SDE, they are able to leverage the properties of differential equations to model the diffusion process more robustly and efficiently.

Despite the advantages, implementing SDEs in diffusion models also introduces new challenges. The primary issue is the computational cost associated with solving SDEs, especially in high-dimensional spaces typical of image data. Additionally, designing and training models to solve these equations requires sophisticated techniques and a deep understanding of both the

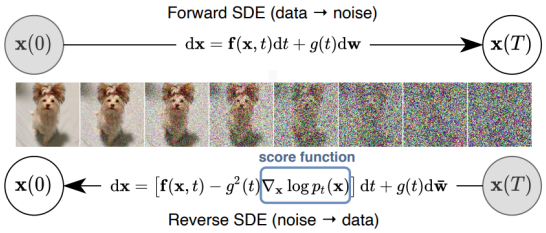


Figure 2.2: Solving a reverse time SDE yields a score-based generative model. Transforming data to a simple noise distribution can be accomplished with a continuous-time SDE. This SDE can be reversed if we know the score of the distribution at each intermediate time step [3]

theoretical and practical aspects of stochastic calculus.

In conclusion, the exploration of SDEs in diffusion probabilistic models represents an exciting advancement, offering a more detailed and continuous approach to modeling the diffusion process. This method holds promise for improving the quality and efficiency of generative models, expanding their applicability in various complex and nuanced tasks.

2.2 Analysis of Competing Products

Denoising Diffusion Probabilistic Models (DDPMs), Variational Autoencoders (VAEs), and Generative Adversarial Networks (GANs) are all prominent types of generative models, each with unique strengths and applications.

2.2.1 Variational Autoencoders (VAE)

Variational Autoencoders (VAEs) have been a significant part of the generative modeling landscape since their introduction by Kingma and Welling in their groundbreaking paper "Auto-Encoding Variational Bayes" in 2013 [4]. VAEs are especially valued for their robustness in learning latent representations and their ability to generate new data points with variations that are meaningful in a learned latent space.

VAEs are particularly celebrated for their ability to learn well-structured latent spaces. This means that similar data points result in close points in the latent space, making these models excellent for tasks that involve data interpolation, anomaly detection, and more. This has been critical for advancements in fields like bioinformatics where understanding the underlying structure of data is crucial.

Early criticisms of VAEs focused on their tendency to produce blurrier samples compared to GANs. However, advancements have been made to address this issue. Techniques such as using hierarchical latent variables, improving the variational bound, or incorporating better decoder architectures have significantly enhanced the quality of the generated samples. For instance, the introduction of VQ-VAE (Vector Quantized VAE) by van den Oord et al. [5] and its successor VQ-VAE-2 by Razavi et al. [6] showcases substantial improvements in the fidelity and diversity of generated images.

VAEs have found applications in a variety of fields beyond just image generation. In drug discovery, VAEs are used to generate novel molecular structures by learning from known chemical compounds. Similarly, in finance, they help in modeling complex financial instruments and in risk management by understanding the distributions of various financial parameters. These applications benefit from VAEs' ability to model complex, high-dimensional data in a compressed, interpretable format.

The flexibility of VAEs allows them to be effectively combined with other neural network architectures, enhancing their utility and performance. For example, the integration of VAEs with Recurrent Neural Networks (RNNs) has led to improved models for sequence data, which is useful in natural language processing and sequential image analysis.

The theoretical aspects of VAEs have also been extensively explored, leading to a deeper understanding and better implementations. Innovations such as the β-VAE, introduced by Higgins et al. [7], offer a way to balance latent channel capacity and independence constraints to control the disentanglement of latent representations, enhancing both interpretability and control over generative models.

2.2.2 Generative Adversarial Networks (GAN)

Generative Adversarial Networks (GANs) have revolutionized the field of generative modeling since their introduction by Ian Goodfellow and his colleagues in 2014 [8]. The unique adversarial training approach of GANs, involving a duel between a generator and a discriminator, has led to substantial advancements in generating high-fidelity and realistic images. The achievements of GANs span various domains, from art creation to medical image analysis.

One of the most significant achievements of GANs is their ability to produce incredibly high-quality and realistic images. For example, the development of models like BigGAN [9] and StyleGAN has pushed the boundaries of how lifelike generated images can be. StyleGAN, in particular, introduced by Karras et al. [10], has become famous for its ability to manipulate and control different aspects of the image generation process, leading to stunningly detailed and controllable outputs. These models demonstrate a leap in the visual quality of generated images, achieving unprecedented resolution and realism.

GANs have been successfully applied to a wide range of applications. In the field of fashion, GANs are used for creating new clothing designs and for virtual try-ons. In video games and virtual reality, they help generate realistic environments and characters. Additionally, GANs have made significant impacts in the medical field, where they are used for data augmentation in medical imaging and for generating synthetic patient data that can be used for training other AI models without privacy concerns.

GANs have also been pivotal in advancing unsupervised and semi-supervised learning techniques. The ability of GANs to learn complex distributions without extensive labeled data has made them valuable for scenarios where labeled data is scarce or expensive to obtain. For instance, models like the Semi-Supervised GAN (SGAN) [11] leverage this capability to improve learning efficiency and accuracy using smaller amounts of labeled data supplemented with larger amounts of unlabeled data.

Despite their potential, GANs were initially known for being difficult to train, with issues such as mode collapse and non-convergence. Over the years, researchers have proposed various techniques to stabilize GAN training, such as introducing new architectures, loss functions, and regularization techniques. The introduction of the Wasserstein GAN (WGAN) [12] marks a significant development in this area, addressing the issue of training stability through a novel loss function that provides a smoother gradient.

The theoretical understanding of GANs has also evolved, leading to more robust and efficient models. Research has delved into the intricacies of GAN dynamics, exploring how different components of the network interact during training. This theoretical exploration has not only improved the practical implementations of GANs but has also contributed to the broader understanding of neural networks and deep learning.

2.3 Specification of Software Standards

- **Google Python Style Guide** is a comprehensive set of guidelines that Google provides to maintain a consistent and high-quality codebase across its Python projects.
- **NumPy** is a fundamental library for scientific computing in Python, providing support for large, multi-dimensional arrays and matrices, along with a large collection of high-level mathematical functions to operate on these arrays.
- **PyTorch** is an open-source machine learning library developed by Facebook, known for its flexibility and ease of use in building and training deep learning models, particularly popular for research and prototyping.

2.4 ResNet Background Theory

Residual Networks represent a seminal advancement in deep learning that emerged to address the vanishing gradient problem occurring in very deep neural networks. ResNet fundamentally altered approaches to network architecture design, enabling the training of networks that are substantially deeper than those previously feasible. This development has had profound implications for the field of computer vision and beyond.

Before ResNet, increasing network depth to improve performance had reached a bottleneck. Typically, deeper networks begin to suffer from the vanishing gradient problem—where gradients become too small to make meaningful adjustments in weights during backpropagation—leading to saturation and then rapid degradation of training accuracy. Moreover, deeper networks often encounter a complexity that makes them computationally expensive and harder to optimize.

ResNet introduced a novel solution to these issues through the use of residual learning. The central hypothesis was that if deeper networks are approximations of shallower ones, then it should be easier to optimize the residual mappings rather than the desired underlying mapping.

The core idea behind ResNet is the introduction of the residual block, which includes shortcut connections (also known as skip connections). These connections allow the input to a layer to be added to its output, effectively forming an identity mapping that skips one or more layers. Mathematically, if $H(x)$ is an underlying mapping to be learned by a few stacked layers, and x is the input, then these layers in ResNet fit another mapping of (2.1). The original function thus becomes (2.2).

$$F(x) := H(x) - x \tag{2.1}$$

$$H(x) = F(x) + x \tag{2.2}$$

This setup makes it easier to optimize the network because learning the residual mapping $F(x)$ is simpler than learning the unreferenced mapping $H(x)$. In practice, this means that even very deep networks can be trained directly through standard backpropagation. This approach addresses both the degradation problem—ensuring that adding more layers does not lead to higher training error—and reduces the impact of the vanishing gradients by providing an alternative pathway for gradient flow during backpropagation.

The standard ResNet models are implemented with varying depths, including ResNet-18, ResNet-34, ResNet-50, ResNet-101, and ResNet-152, which indicate the number of layers. Each ResNet block is a composition of several convolutional layers, batch normalization layers, and ReLU activations. In deeper ResNet versions (50, 101, 152), bottleneck layers are introduced

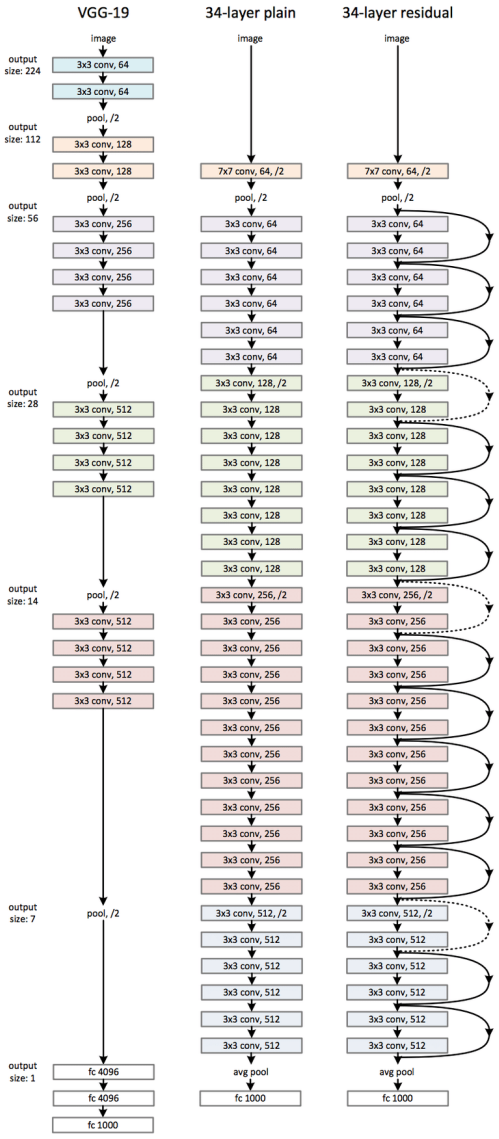


Figure 2.3: Example network architectures for ImageNet. **Left**: the VGG-19 model [13] (19.6 billion FLOPs) as a reference. **Middle**: a plain network with 34 parameter layers (3.6 billion FLOPs). **Right**: a residual network with 34 parameter layers (3.6 billion FLOPs). The dotted shortcuts increase dimensions. [1]

to reduce the model size and complexity. Each convolutional layer within these bottlenecks is tightly packed with 1×1 and 3×3 convolutions.

2.4.1 Plain Network

The plain network architecture referenced in the context of ResNet typically comprises stacked convolutional layers, where each layer is followed by batch normalization and a ReLU activation function (Figure 2.3, **Left**). This architecture builds on the basic principles of CNNs, which are designed to capture hierarchical patterns in image data through a series of filtering and pooling operations.

In a plain network, convolutional layers form the core components. These layers apply a set of learnable filters to the input image, capturing various features at different levels of abstraction. For example, the initial layers might capture simple features like edges and textures, while deeper layers capture more complex patterns such as object parts.

Following each convolutional operation, batch normalization is applied. This technique normalizes the input layer by adjusting and scaling the activations, helping to maintain mean output close to 0 and the output standard deviation close to 1. This normalization helps to combat issues in training deep networks by stabilizing the learning process and reducing the number of training epochs required.

After batch normalization, a non-linear activation function—typically the Rectified Linear Unit (ReLU)—is used. The ReLU function introduces non-linear properties to the system, allowing the network to learn more complex patterns. It is defined as $f(x) = max(0, x)$, which has been shown to accelerate the convergence of stochastic gradient descent compared to traditional sigmoid or tanh functions.

The architecture often includes pooling layers, which reduce the spatial size of the representation, diminishing the amount of parameters and computation in the network. Pooling helps in making the detection of features invariant to scale and orientation changes.

Towards the end of the network, fully connected layers are typically used to map the learned features to the desired output size. For instance, in classification tasks, these layers might culminate in a softmax activation function that distributes the output into probability scores corresponding to various classes.

The plain architecture, while foundational, encounters significant challenges as network depth increases:

- **Vanishing/Exploding Gradients**: In very deep networks, gradients propagated back through the network can vanish (become very small) or explode (become very large), which significantly hampers the ability to learn.
- **Degradation Problem**: As depth increases, network accuracy gets saturated and then degrades rapidly. Interestingly, this degradation is not due to overfitting, and adding more layers leads to higher training error.

The plain network architecture is crucial for understanding the baseline upon which ResNet improves. By addressing the shortcomings of the plain networks, especially the degradation problem through the introduction of skip connections (or shortcut connections), ResNet allows much deeper networks to be trained effectively. This capability to train deeper networks without a drop in performance has set new benchmarks in various machine learning tasks and underscores the transformative impact of ResNet in the landscape of deep learning.

2.4.2 Residual Network

The cornerstone of the ResNet architecture is the concept of residual learning. As networks deepen, training becomes increasingly difficult due to the vanishing gradient problem, where gradients of the network's layers diminish as they propagate back through the network during training. Traditional networks attempt to learn the desired underlying mapping directly. In contrast, ResNet reformulates the layers as learning residual functions with reference to the layer inputs, thereby simplifying the learning process (Figure 2.3, **Right**).

A typical ResNet is composed of several stacked residual blocks. Each residual block has two main pathways:

- **Identity Shortcut Connection**: This pathway allows the input to bypass the main layers of the block and connect directly to the output of the block, adding the input x to the output of the block's layers $F(x)$. This direct path forwards the input without any alteration, preserving the strength of the gradient as it back-propagates through the network.
- **Weighted Layers**: Typically consisting of two or three convolutional layers (depending on the variant of ResNet), each followed by batch normalization and a ReLU activation function. These layers aim to learn the residual functions, denoted as $F(x)$, where x is the input to the layers.

The output of a residual block is then (2.2), where $F(x)$ represents the learned residual function and x is the identity mapping. The addition of $F(x)$ and x is element-wise and may involve dimensionality adjustments (using 1×1 convolutions) if necessary to match the dimensions.

ResNet comes in various depths, which are generally tailored to the complexity of the tasks they are intended to perform. Common variants include:

- **ResNet-18** and **ResNet-34**: These versions use blocks with two convolutional layers each.
- **ResNet-50**, **ResNet-101**, and **ResNet-152**: These deeper models utilize bottleneck blocks designed to reduce the model size and computational load. A bottleneck block contains three layers: a 1×1 convolution that reduces the dimension, a 3×3 convolution, and another 1×1 convolution that restores the dimension.

In this project the denoising network is not one of the standard ImageNet classifiers but a residual UNet (ResUNet) assembled from the basic two-layer residual block described above. Bottleneck blocks of the kind used in ResNet-50 are designed for the deep, heavily downsampled feature hierarchies demanded by 224×224 ImageNet classification; at the 32×32 resolution of CIFAR-10 the spatial extent is exhausted after only a few downsampling stages, so the additional 1×1 compression they provide yields little benefit for a considerable increase in implementation complexity. A shallower network built from basic blocks is therefore adopted, and the classification head is replaced by an encoder-decoder structure so that the network maps an image to an image rather than to a class score, as a denoising network must.

The resulting architecture retains two downsampling stages at the 32×32 resolution of CIFAR-10. At the 64×64 resolution of CelebA a third stage is inserted and the channel widths are narrowed accordingly, so that the two configurations remain comparable in size.

- **Encoder stage 1** maps the three input channels to 64 feature channels at 32×32, after which max pooling halves the resolution.
- **Encoder stage 2** maps 64 channels to 128 at 16×16, followed by a second max pooling to 8×8.

- A **residual bridge** operates on the 128 channels at the coarsest resolution.
- **Decoder stage 2** upsamples to 16 × 16 by transposed convolution and concatenates the stage-2 encoder features, giving 256 channels.
- **Decoder stage 1** upsamples to 32 × 32 and concatenates the stage-1 encoder features, giving 128 channels.
- A final 1 × 1 convolution projects the 128 channels back to the three image channels.

Each residual block follows the two-layer form used in ResNet-18 and ResNet-34: a 3 × 3 convolution, a normalisation layer, a non-linear activation, a second 3 × 3 convolution and a further normalisation, with the block input added to the result. Group normalisation is used in place of the batch normalisation of the original ResNet. The running statistics that batch normalisation accumulates would be estimated over clean images, whereas the input actually presented to a denoising network is the noised image x_t, whose distribution changes with the timestep; group normalisation avoids this mismatch by normalising within each example. Because the channel count is preserved within a block, the shortcut is a pure identity mapping and no 1 × 1 projection is required. The connections between corresponding encoder and decoder stages are concatenations rather than additions, which is the convention inherited from the UNet family and is what allows the fine spatial detail lost during downsampling to be recovered in the decoder.

2.4.3 Reference Denoising Architecture

The network proposed above is assessed against the denoising architecture introduced with the DDPM itself [14], which serves as the reference design throughout this work. It is also a UNet built from residual blocks, but differs from the network of the previous section in four respects.

- It uses **four resolution stages** rather than two or three, so the coarsest feature map is reached through a deeper hierarchy.
- **Self-attention** is inserted at the 16×16 resolution and at the bottleneck, giving each spatial position access to the whole feature map rather than only to its convolutional neighbourhood.
- A **skip connection is taken from every residual block**, not one per resolution stage, so the decoder receives a considerably finer record of the encoder activations.
- Resolution is changed by **strided convolution and nearest-neighbour upsampling followed by convolution**, in place of max pooling and transposed convolution.

Because the channel count varies within a stage, the shortcut in these blocks cannot be a pure identity mapping and a 1×1 projection is applied where the dimensions differ. The configuration published for CIFAR-10 uses a base width of 128 channels and amounts to roughly 35 million parameters, which is more than an order of magnitude larger than the network proposed here. Comparing the two at their published sizes would confound architecture with capacity, so the base width is reduced until the parameter counts agree to within a few per cent; the comparison then reflects the arrangement of the layers rather than the number of them.

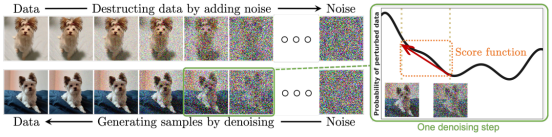


Figure 2.4: Diffusion models smoothly perturb data by adding noise, then reverse this process to generate new data from noise. Each denoising step in the reverse process typically requires estimating the score function (see the illustrative figure on the right), which is a gradient pointing to the directions of data with higher likelihood and less noise. [15]

2.5 Diffusion Model Background Theory

Diffusion models (Figure 2.4) operate by modeling the data generation process as a Markov chain of gradual, stepwise transformations from a data distribution into a predetermined noise distribution, typically Gaussian noise. This process is characterized mathematically by a series of forward diffusion steps that progressively add noise to the data until only noise remains.

2.5.1 Forward Process

The forward process (or the diffusion process) can be described by the following stochastic differential equation (SDE):

$$dX_t = -\frac{1}{2}\beta(t)X_t dt + \sqrt{\beta(t)}dW_t \tag{2.3}$$

Here, X_t represents the data at time t, $\beta(t)$ is a variance schedule that governs the noise level at each step, and dW_t denotes the increment of a Wiener process, reflecting the stochastic nature of the noise addition. The parameter $\beta(t)$ typically increases with t, indicating that the amount of noise added increases over time.

2.5.2 Reverse Process

The reverse process aims to reconstruct the original data from the noise. This is where diffusion models utilize the concept of denoising. In the reverse process, the model learns to estimate the gradients of the log probability density of the data with respect to the data itself, given the noisy data at each step. This gradient is used to perform a step-by-step denoising through the following approximate reverse SDE:

$$dX_t = \left[-\frac{1}{2}\beta(t)X_t - \beta(t)\nabla_{X_t}\log p_t(X_t)\right]dt + \sqrt{\beta(t)}\mathrm{d}\bar{W}_t \tag{2.4}$$

The term $\nabla_{X_t}\log p_t(X_t)$ represents the score function that the model learns; this function guides the reverse diffusion process to effectively recover the data distribution from the noise.

2.5.3 Scheduling

The forward process in a diffusion model gradually adds noise to the original data over a series of steps. This is often done in a controlled manner according to a predefined noise schedule. The noise schedule determines the amount of noise added at each step and is typically parameterized by a variance schedule $\beta(t)$, where t denotes the timestep.

The reverse process involves denoising the data. This process uses the learned parameters to gradually remove the noise from the data, aiming to revert to the original data distribution. The reverse process follows a reverse noise schedule, which ideally mirrors the forward noise

schedule but in reverse order. The model learns to predict the clean data from the noisy data step-by-step, effectively learning the conditional distribution of the cleaner data given the noisier data.

The optimization of the noise schedule is a key area of research in diffusion models. The schedule needs to be carefully designed to balance the trade-offs between the model's stability, sampling speed, and quality of generated samples. Researchers often experiment with different forms of the noise schedule, such as linear, cosine, or learned schedules, each providing different benefits:

- **Linear Schedules** increment noise linearly over time and are straightforward but may not always capture the nuances of more complex data distributions.
- **Cosine Schedules** vary noise according to a cosine function, often leading to smoother transitions in noise levels.
- **Learned Schedules** involve learning the noise schedule directly from the data, allowing the model to adaptively determine the best way to add and remove noise based on the specific characteristics of the data.

In this project, we will explore and compare various scheduling methods to determine the optimal noise schedule for the Denoising Diffusion Probabilistic Model (DDPM). The investigation will involve systematically implementing these different scheduling strategies to assess their impact on the model's performance. By analyzing the effects of these schedules on the fidelity, stability, and efficiency of the generated outputs, the study aims to identify the most effective scheduling method tailored to the specific characteristics and requirements of the DDPM. This comparative analysis will not only enhance our understanding of the model's dynamics but also contribute to optimizing its generative capabilities in practical applications.

3 Results

3.1 Experimental Configuration

The six runs specified in Section 1.1.2 were carried out on a single NVIDIA GeForce RTX 4070. Every run used $T = 1000$ diffusion steps under the cosine schedule, a batch size of 128, the Adam optimiser at a learning rate of 2×10^{-4}, thirty epochs, random horizontal flipping as the only augmentation, and the same random seed. Training used bfloat16 mixed precision with the channels-last memory format; because these settings alter neither the model definition nor the objective, they affect the wall-clock cost of a run but not the comparison between architectures.

Table 3.1 records the resulting parameter budgets. The proposed network and the plain variant are identical in size, as an identity shortcut introduces no parameters, so the comparison between them isolates residual learning as the only variable. The reference network was narrowed until it fell within ten per cent of the proposed network; it is in fact the smallest of the three, which means that any advantage it shows cannot be attributed to additional capacity.

Table 3.1: Parameter counts. The proposed and plain networks are identical in size by construction; the reference network is the smallest of the three.

Dataset	Architecture	Parameters	Relative to ResUNet
CIFAR-10	ResUNet	2 473 475	—
CIFAR-10	plain UNet	2 473 475	100.0%
CIFAR-10	DDPM UNet	2 356 739	95.3%
CelebA	ResUNet	2 612 419	—
CelebA	plain UNet	2 612 419	100.0%
CelebA	DDPM UNet	2 356 739	90.2%

3.2 Training Convergence

Table 3.2 reports the epoch-averaged noise-prediction loss at five points during training. Two patterns are visible, and they point in opposite directions.

Table 3.2: Epoch-averaged training loss. Lower is better. The proposed network leads at the first epoch on both datasets and is overtaken by the fifth.

		Epoch				
Dataset	Architecture	1	5	10	20	30
CIFAR-10	ResUNet	**0.13775**	0.07111	0.06619	0.06249	0.06108
CIFAR-10	plain UNet	0.14633	**0.07033**	**0.06555**	**0.06212**	0.06084
CIFAR-10	DDPM UNet	0.18508	0.07135	0.06593	0.06165	**0.05967**
CelebA	ResUNet	**0.14205**	0.04735	0.04110	0.03844	0.03716
CelebA	plain UNet	0.16413	0.04705	0.04055	0.03799	0.03679
CelebA	DDPM UNet	0.15588	**0.04316**	**0.03779**	**0.03573**	**0.03476**

The first is an early advantage for residual learning. After a single epoch the proposed network has the lowest loss on both datasets, ahead of the plain variant by 5.9% on CIFAR-10 and by 13.5% on CelebA. Since the two networks are identical apart from the identity shortcut, this difference is attributable to that shortcut alone, and it is consistent with the argument of the original ResNet work that a residual formulation is easier to optimise than an unreferenced one.

The second pattern is that the advantage does not survive. By the fifth epoch the plain network has already overtaken the proposed one on both datasets, and at convergence it remains marginally ahead: 0.06084 against 0.06108 on CIFAR-10 and 0.03679 against 0.03716 on CelebA. These final gaps are 0.4% and 1.0% respectively. Section 3.4 shows that this near-equality on the training objective is profoundly misleading about the quality of the images the two networks actually generate.

The reference network behaves differently again. It begins worst on CIFAR-10, at 0.18508, which is what a deeper network with more layers to coordinate would be expected to do, but it overtakes both competitors by the twentieth epoch and finishes with the lowest loss on both datasets.

3.3 Restoration Accuracy

The restoration diagnostics corrupt a held-out test image to a chosen timestep and measure how completely the network recovers it in a single step. Table 3.3 reports both measures over a thousand test images at six noise levels.

Table 3.3: Restoration accuracy against noise level. SSIM (higher better) above, PSNR in decibels (higher better) below. Differences between the proposed and plain networks are negligible throughout; the reference network leads at every level.

	CIFAR-10			CelebA		
t	ResUNet	plain	DDPM UNet	ResUNet	plain	DDPM UNet
SSIM						
50	0.9712	0.9713	**0.9713**	0.9708	0.9710	**0.9727**
100	0.9388	0.9393	**0.9395**	0.9444	0.9451	**0.9485**
200	0.8701	0.8707	**0.8740**	0.8947	0.8953	**0.9027**
400	0.7286	0.7281	**0.7382**	0.7996	0.8004	**0.8153**
600	0.5681	0.5694	**0.5826**	0.6891	0.6894	**0.7154**
800	0.3529	0.3438	**0.3642**	0.5117	0.5023	**0.5510**
Mean	0.7383	0.7371	**0.7450**	0.8017	0.8006	**0.8176**
PSNR (dB)						
50	33.50	33.52	**33.54**	35.51	35.54	**35.77**
100	29.89	29.91	**29.95**	32.20	32.24	**32.49**
200	26.21	26.23	**26.34**	28.80	28.83	**29.10**
400	22.35	22.37	**22.52**	25.07	25.10	**25.40**
600	19.61	19.61	**19.80**	22.23	22.25	**22.61**
800	16.63	16.50	**16.87**	18.82	18.74	**19.18**
Mean	24.70	24.69	**24.84**	27.10	27.12	**27.43**

Three observations follow. The first is that at $t = 50$ the three networks are indistinguishable to four decimal places on SSIM and to within 0.04 dB on PSNR. An image at that noise level is barely corrupted, and almost any denoiser recovers it; the measurement discriminates only as the corruption grows.

The second is that the proposed and plain networks cannot be separated. Their mean SSIM differs by 0.0012 on CIFAR-10 and 0.0011 on CelebA, and their mean PSNR by 0.01 dB and 0.02 dB. To put those figures in context, the same evaluation repeated in a separate session, with the model weights unchanged but the injected noise redrawn, moved the CIFAR-10 mean SSIM of the proposed network from 0.7388 to 0.7383 and that of the plain network from 0.7373 to 0.7371. The run-to-run variation is therefore of the same order as the difference between the two networks, and at $t = 800$ the plain network's value moved by 0.0055 between sessions against a 0.0091 gap to its competitor. No claim about the identity shortcut can be supported at this resolution.

The third is that the reference network leads at every noise level on both datasets and on both measures, and its margin grows with t. Its advantage is consistently larger on CelebA than on CIFAR-10, which is the pattern the higher resolution was introduced to expose.

3.4 Distribution-Level Quality

The measures reported so far compare the networks as denoisers. Table 3.4 compares them as generators, using two thousand samples drawn from each model through the complete thousand-step reverse process.

Table 3.4: Distribution-level quality from 2048 samples per model. FID lower is better, IS higher is better. The Inception Score is reported for CIFAR-10 only, since CelebA is a single semantic category and the measure is undefined without class diversity.

Dataset	Architecture	FID	IS	Mean SSIM
CIFAR-10	ResUNet	94.93	4.711	0.7383
CIFAR-10	plain UNet	84.29	4.625	0.7371
CIFAR-10	DDPM UNet	**68.09**	**5.655**	**0.7450**
CelebA	ResUNet	**76.06**	—	0.8017
CelebA	plain UNet	135.18	—	0.8006
CelebA	DDPM UNet	76.93	—	**0.8176**

This table contains the most important result of the project, and it is not the one the restoration measures predicted.

On CelebA the proposed network attains an FID of 76.06 against 135.18 for the plain variant. The plain network is 78% worse. Yet these two networks have identical parameter counts, identical computational cost, a final training loss that favours the plain network by one per cent, and mean SSIM and PSNR values that agree to within 0.0011 and 0.02 dB. **Two networks that denoise equally well generate images of very different quality.** The proposed network is also marginally ahead of the reference network here, at 76.06 against 76.93, the only measurement anywhere in this work on which it leads.

On CIFAR-10 the ordering is the other way round: the plain network attains 84.29 against 94.93 for the proposed one, an 11% advantage, and the reference network leads both at 68.09. The Inception Score agrees with FID on the reference network's lead and separates the proposed and plain networks by only 0.086, which is within the reported standard deviation of either.

The natural explanation for the reversal is depth. At 32×32 the proposed network has two resolution stages; at 64×64 it has three. Residual shortcuts exist to make deep networks optimisable, and the deeper configuration is where they would be expected to matter. On this reading the two-stage network is shallow enough that the shortcut is unnecessary and its constraint on

the function class costs a little, while the three-stage network is deep enough to benefit substantially. A single seed per configuration cannot establish this, but the effect on CelebA is far too large to be attributed to run-to-run variation of the kind quantified in the previous section, and it reproduced in an earlier evaluation at a smaller sample size, where the same three models scored 117.83, 184.48 and 115.12.

The deeper methodological point is that the restoration measures were unable to anticipate any of this. The training objective averages the noise-prediction error uniformly over all timesteps, whereas generation composes a thousand such predictions in sequence. Errors that are small on average but systematic at particular timesteps compound through that composition. A network can therefore be an excellent denoiser and a poor generator, and only a distribution-level measure detects the difference.

3.5 Memorisation

Because the CelebA samples are recognisable faces, it is necessary to establish that the models are generating rather than reproducing. Table 3.5 reports the mean distance from each generated sample to its nearest training image, against two controls.

Table 3.5: Mean pixel-space L_2 distance to the nearest training image. A generated sample that is no closer than a blurred real image gives no evidence of reproduction.

	CIFAR-10			CelebA		
	ResUNet	plain	DDPM	ResUNet	plain	DDPM
Generated	18.00	16.93	17.06	43.94	47.02	43.56
Real held-out images		18.68			38.41	
Blurred held-out		16.09			35.97	

On CelebA every model produces samples that are *further* from the training set than genuine unseen photographs are, by between thirteen and twenty-two per cent, which settles the question directly.

CIFAR-10 requires the second control. The generated samples sit slightly closer to the training set than real held-out images do, at between 0.91 and 0.96 of the reference distance, and taken alone that could be read as mild reproduction. Blurring a real held-out image with a 3×3 mean filter, however, reduces its distance to 16.09, or 0.86 of the reference. Smoothing an image moves it towards the mean of the data and therefore towards everything in it. All three models remain further away than that purely smoothing-induced floor, so the shortfall is explained by the softness of samples from an undertrained model rather than by copying.

Figure 3.1 shows the eight closest generated and training pairs for the proposed network on CelebA. The nearest training image is a different person in every case.

3.6 Computational Cost

Table 3.6 records the steady-state time for one training epoch. The proposed and plain networks are within two per cent of one another on both datasets, confirming in practice what the parameter counts assert in principle: removing an identity shortcut changes neither the parameter count nor the arithmetic performed.

The reference network costs 2.4 times as much per epoch on CIFAR-10 and 2.2 times as much on CelebA, despite having fewer parameters than either competitor. The cost is incurred by depth and by attention rather than by weight count. Its advantage on CIFAR-10 is decisive and

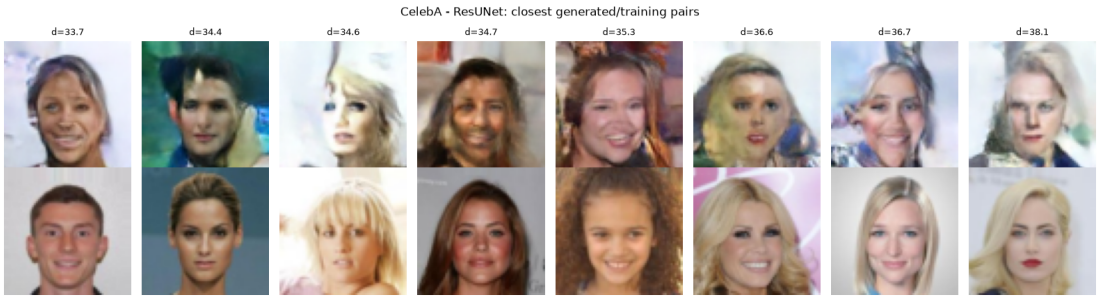


Figure 3.1: The eight CelebA samples closest to the training set (top) with their nearest training image (bottom), for the proposed network. No pair depicts the same person.

Table 3.6: Steady-state training time per epoch. The first epoch of each dataset absorbs convolution autotuning and is excluded.

Dataset	ResUNet	plain UNet	DDPM UNet
CIFAR-10 (32 × 32)	16.0 s	15.8 s	38.5 s
CelebA (64 × 64)	32.0 s	31.4 s	68.3 s

worth the price; on CelebA it is matched by the proposed network at less than half the training cost.

3.7 Generated Samples

3.7.1 A Note on the Reverse Process

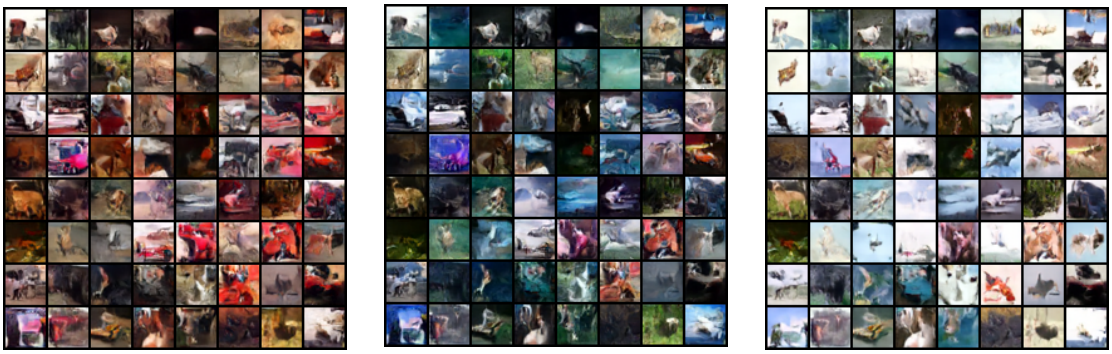
The first attempt at unconditional sampling produced saturated noise from every model. The cause was not the models but the implementation of the reverse step. Writing the posterior mean directly as $\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\varepsilon_\theta\right)/\sqrt{\alpha_t}$ is algebraically correct only while the implied estimate of x_0 remains inside the data range. The factor $1/\sqrt{\alpha_t}$ reaches 31.6 at $t = 999$ under the clipped cosine schedule, so any error in ε_θ is amplified by that factor and accumulates over the thousand steps of the reverse process. Tracing the sample statistics confirmed the effect: the standard deviation of x grew from 1.0 at the start of the trajectory to approximately 16 by its end, and over 95% of pixels finished outside the valid range. The same divergence occurred in single precision, which excludes reduced precision as the cause.

The remedy is the one used in the reference implementation: recover x_0 explicitly, clip it to $[-1, 1]$, and rebuild the posterior mean from the clipped estimate. With this correction the terminal standard deviation falls to between 0.48 and 0.64 across the six models, matching the scale of the training data, and no pixel leaves the valid range. All samples reported here use the corrected procedure. The episode illustrates a property of the reverse process that the forward derivation does not make obvious: the algebraic form of the posterior mean is stable only for a network that is already accurate, and the clipping step is what makes sampling robust to a network that is not.

3.7.2 Samples

Figure 3.2 and Figure 3.3 show sixty-four samples drawn from each model. Every model was given the same initial noise, so differences between the panels are attributable to the denoising

network.



ResUNet plain UNet DDPM UNet

Figure 3.2: Unconditional CIFAR-10 samples after thirty epochs, all three models starting from identical noise.

ResUNet plain UNet DDPM UNet

Figure 3.3: Unconditional CelebA samples at 64×64 after thirty epochs, all three models starting from identical noise.

On CIFAR-10 all three models produce coherent local texture and plausible figure-ground separation, and some samples suggest recognisable categories such as animals or vehicles, but none is identifiable with confidence. This is the expected outcome for networks of two and a half million parameters trained for eleven thousand optimisation steps; the published CIFAR-10 configuration is roughly fourteen times larger and is trained for two orders of magnitude longer.

The CelebA panels are where the FID result becomes visible. All three models generate face-like images with correct global arrangement, but the plain network's faces are noticeably less coherent, with more frequent failures of facial structure, which is consistent with its FID of 135.18 against 76.06 and 76.93 for the other two. The proposed and reference networks are difficult to separate by eye, as their almost identical scores would suggest. Colour saturation is exaggerated in all three, a known consequence of clipping the estimate of x_0 at every step of an undertrained reverse process.

Figure 3.4 shows the reverse trajectory of the proposed network on both datasets, sampled every twenty steps. Structure emerges late: the first two thirds of the trajectory refine what remains essentially noise, and the recognisable image forms over the final few hundred steps.

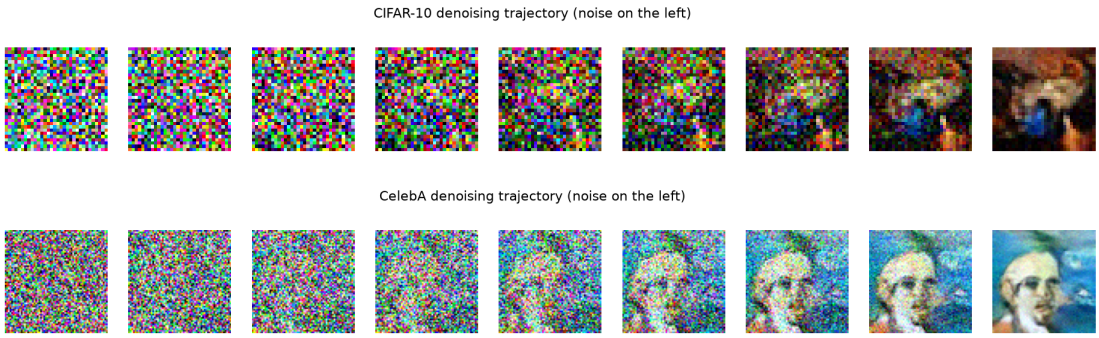


Figure 3.4: Reverse trajectories of the proposed network on CIFAR-10 (top) and CelebA (bottom), with pure noise on the left and the final sample on the right.

3.8 Summary

Figure 3.5 collects the training curves and the restoration measurements. Four findings follow from the experiment.

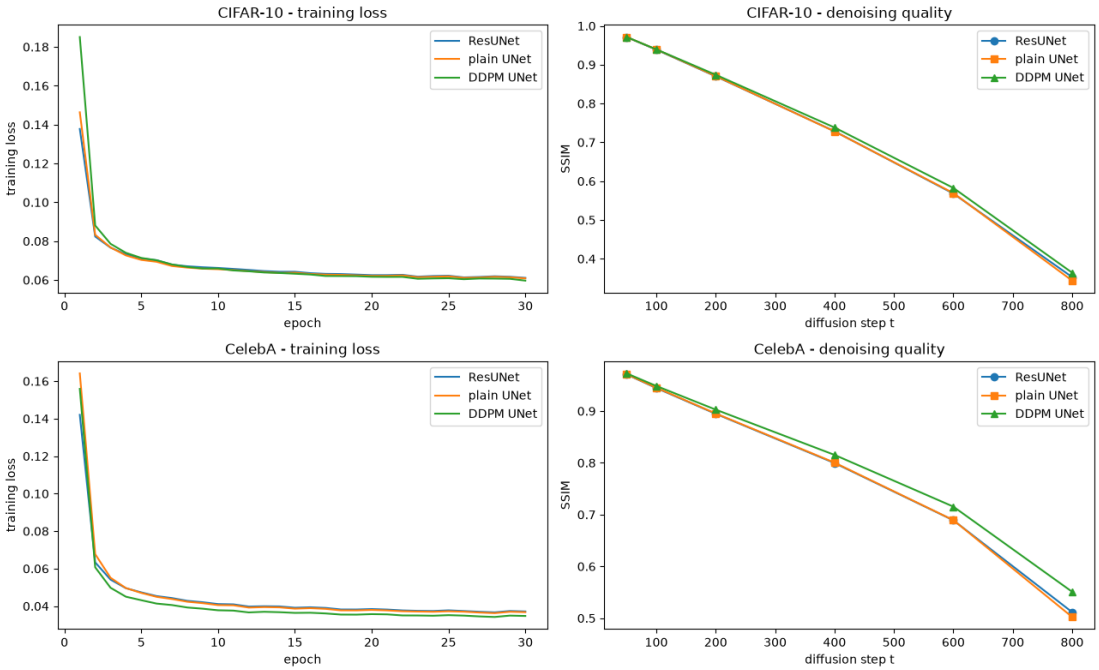


Figure 3.5: Training loss (left) and denoising accuracy against noise level (right) for CIFAR-10 (top) and CelebA (bottom).

- **Residual shortcuts accelerate early optimisation.** The proposed network reaches a lower loss than its shortcut-free twin after a single epoch, by 5.9% on CIFAR-10 and 13.5% on CelebA. As the two networks are otherwise identical, the shortcut is the only possible cause.
- **Restoration measures cannot separate the two networks.** Their final training loss, mean SSIM and mean PSNR agree to within one per cent, half of one per cent and 0.02 dB respectively, and the run-to-run variation of the evaluation is of the same order as the differences themselves.

- **Generative quality separates them sharply, and the direction depends on depth.** At 64×64, where the proposed network has three resolution stages, it attains an FID of 76.06 against 135.18 without the shortcut. At 32×32, with two stages, the ordering reverses to 94.93 against 84.29. Networks that denoise identically well do not generate equally well, and no restoration measure predicted this.
- **The reference architecture remains the strongest overall.** It leads on every restoration measure on both datasets and on CIFAR-10 FID by a wide margin, using fewer parameters than either competitor, at roughly 2.3 times the training cost. Only on CelebA FID is it matched, by the proposed network at less than half the cost.

The hypothesis stated in Chapter 1 therefore receives partial and conditional support. The combination of residual learning with a UNet backbone does not improve the quality of generated images in general, and at the lower resolution it is mildly harmful. At the higher resolution, where the network is deep enough for the optimisation difficulty that residual learning was invented to address, the improvement is substantial and it is invisible to every measure except the distribution-level one.

Conclusions

This project set out to determine whether combining residual learning with the UNet backbone conventionally used as the denoising network of a diffusion model improves the quality of the images it generates. A Denoising Diffusion Probabilistic Model was implemented from first principles and the diffusion framework was held fixed while three denoising networks were substituted into it: the proposed residual UNet, an otherwise identical network with the identity shortcut removed, and the reference architecture of Ho et al. narrowed to a matching parameter budget. Each was trained on CIFAR-10 at 32×32 and on CelebA at 64×64 under identical conditions.

Findings

The answer to the question as posed is that residual learning helps, but conditionally, and not in the way the project anticipated.

The shortcut makes optimisation easier at the outset. After one epoch the proposed network leads its shortcut-free twin by 5.9% on CIFAR-10 and 13.5% on CelebA, and since the two differ in nothing else the shortcut is the only possible cause. That advantage is gone by the fifth epoch and does not reappear: at convergence the two networks are indistinguishable on the training objective, on structural similarity and on peak signal-to-noise ratio, with all differences at or below the run-to-run variation of the evaluation itself.

They are not, however, indistinguishable as generators. On CelebA the proposed network attains a Fréchet Inception Distance of 76.06 where the plain variant reaches only 135.18, and on CIFAR-10 the ordering reverses to 94.93 against 84.29. The most plausible reading is that the effect tracks depth: the CelebA configuration has three resolution stages against two for CIFAR-10, and residual shortcuts exist precisely to make deeper networks optimisable. Where the network is shallow enough not to need one, the constraint it imposes costs a little; where it is deep enough to struggle without one, it helps a great deal.

The reference architecture remains the strongest network overall. It leads on every restoration measure on both datasets and on CIFAR-10 FID by a wide margin, while using fewer parameters than either competitor, at roughly 2.3 times the training cost per epoch. Only on CelebA FID is it matched, and there the proposed network achieves the same quality for less than half the compute.

Two results of a different kind emerged from the work and are worth recording. The first is methodological: a network can denoise as well as another and generate images that are 78% worse by FID. The training objective averages the noise-prediction error uniformly over timesteps, whereas generation composes a thousand such predictions in sequence, so errors that are small on average but systematic at particular timesteps compound through that composition. Evaluating a diffusion model by reconstruction alone is not sufficient. The second is practical: the algebraic form of the posterior mean used in the reverse process is numerically stable only for a network whose noise prediction is already accurate. Without clipping the implied estimate of x_0 at every step, the reverse process diverged from a standard deviation of 1 to approximately 16, in single precision as well as mixed, and produced saturated noise from every model.

Limitations

Each configuration was trained once, with a single random seed. The run-to-run variation of the evaluation was measured and is small relative to the FID differences reported, but it is not a substitute for repeated runs, and no confidence interval can be attached to any figure here. The depth explanation offered for the reversal between datasets is consistent with the evidence but is not established by it; separating depth from resolution would require training the two-stage and three-stage configurations on the same images.

Thirty epochs amounts to roughly eleven thousand optimisation steps, some two orders of magnitude short of the schedule used for the published CIFAR-10 configuration, and the networks are about fourteen times smaller. The absolute FID values are correspondingly poor and should not be compared with the literature; only the relative ordering between the three networks, which were trained identically, carries meaning. The Fréchet Inception Distance was computed from two thousand samples, which is the smallest number at which the estimate is not degenerate given the dimensionality of the Inception feature space, and remains biased upward at that size.

The memorisation check operates in pixel space and would not detect a reproduction that had been shifted or recoloured. Negative Log-Likelihood was excluded on the grounds that this implementation fixes the reverse-process variances rather than learning them. The comparison against Variational Autoencoders and Generative Adversarial Networks set out in Chapter 2 was not carried out.

Further work

The immediate priority is repetition. Three seeds per configuration would establish whether the FID differences reported here are stable, and would cost no more than the original experiment.

The depth hypothesis is directly testable. Training two-stage, three-stage and four-stage variants of the proposed network on a single dataset, each with and without the shortcut, would isolate depth from resolution and determine whether the benefit of residual learning grows with the former as the argument predicts.

Beyond that, learning the reverse-process variances rather than fixing them would make the variational bound informative and bring Negative Log-Likelihood within reach, and the Variational Autoencoder and Generative Adversarial Network baselines would place the diffusion results in the wider context that Chapter 2 describes but does not measure.

A Appendicies

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